Last updated in Sept 2026

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Software & Data Systems · Technical Project Delivery · AI & Automation

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Summary

Computational scientist (PhD) who builds software and hardware systems end to end — defining requirements and acceptance criteria up front, implementing and validating the solution, then documenting and training non-technical users to run it independently. Nine years building research software in Python, with Rust for real-time components, across data pipelines, HPC infrastructure, computer vision and applied AI. Native English speaker; German B1–B2.

Experience

Doctoral Researcher — Research Software & Systems Engineering, MPI for Neurobiology of Behavior / University of Bonn – Bonn, Germany — 2019 – 2025

- Designed and built a real-time distributed measurement system end to end — defined datasets, test setups and acceptance criteria, then implemented multi-camera acquisition, real-time 3D tracking and hardware triggering at 10 ms closed-loop latency (applied to freely flying insects).
- Ran the technology evaluation for the timing-critical layer: benchmarked the Python implementation against the latency requirements, found it insufficient, and rewrote acquisition and triggering in Rust; kept Python for surrounding tooling and analysis.
- Built reproducible data pipelines processing hundreds of recordings in HDF5/Parquet, run in parallel locally and as Slurm batch jobs on HPC; open-sourced the system as OptoFly (github.com/mpinb/optofly, GPL-3.0).
- Owned the full lifecycle — requirements, build, validation, documentation, handover and support — as the interface between non-technical users and the implementation; trained colleagues to run it unsupervised; supervised one MSc student.
- **Result:** first-author paper in Current Biology (2025); the system remains in daily use as one of the lab's standard tools.

MSc Researcher — Scientific Software & Data Analysis, Technion – Israel Institute of Technology – Haifa, Israel — 2016 – 2018

- Built MATLAB signal-processing and statistics pipelines for large datasets; modernised and documented the lab's legacy analysis codebase, which was then adopted lab-wide, and trained incoming researchers on it. Also taught Python, statistics and ML to non-programmer medical students (teaching assistant). Co-first-author paper in Current Biology (2021).

Selected Projects

Automated visual inspection pipeline — MPI for Neurobiology of Behavior – Bonn, Germany — 2025 – 2026

- Built imaging hardware and an OpenCV/FFmpeg + YOLO pipeline to replace a manual visual scoring process; hardware, acquisition and detection ran end to end within a three-month contract, classification validated at notebook stage.

Skills

AI & automation: Claude Code, MCP-based tool/skill integration, agentic coding harnesses (opencode, pi), OpenAI-compatible APIs, OpenRouter; computer-vision models for automating manual processes

Software, data & infrastructure: Python (primary), Rust, MATLAB, R, Bash, SQL, Git/GitHub, NumPy/pandas/scikit-learn, HDF5/Parquet, Linux, Docker, HPC (Slurm)

Computer vision & hardware: OpenCV, FFmpeg, camera calibration, object detection (YOLO), closed-loop real-time instrumentation

Working methods & tools: Requirements analysis and acceptance criteria; full software lifecycle from build to handover and support; iterative, incrementally validated development; structured written analyses and recommendations; Microsoft 365

Languages: English (native), Hebrew (native), German (B1–B2, actively studying)

Education

University of Bonn / MPI for Neurobiology of Behavior, PhD in Neuroscience — computational (magna cum laude) – Bonn, Germany — 2019 – 2025

Technion – Israel Institute of Technology, MSc in Neuroscience (summa cum laude) – Haifa, Israel — 2016 – 2018

University of Haifa, BSc in Biology and Psychology – Haifa, Israel — 2013 – 2016